Modular Symmetry of $S[\Theta]$ and the Derivation of $R_\psi = 1$

Two independent paths to the self-dual point

Ing. David Jaroš

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Abstract

We derive $R_\psi = 1$ (self-dual point $\tau_E = i$) via two independent paths.

Path 1 — Casimir polynomial [L0]+[MC]. If the effective action S_{eff} is invariant under $R_\psi \to 1/R_\psi$, combining the S-invariance condition with stationarity yields the polynomial $R_\psi^6 + R_\psi^4 + R_\psi^2 = 3$, whose unique positive solution is $R_\psi = 1$. N_{eff} cancels; the result holds for any N_{eff}. The S-invariance of S_{eff} is assumed [[**MC**]]; the polynomial itself is an algebraic identity [[**L0**]].

Path 2 — Partition function [L1] (conditional). The one-loop partition function of the UBT ψ-sector is $Z[\tau_E] = \eta(\tau_E)^{-12}$, derived from $\det P_{\text{eff}} = \eta(\tau_E)^2$ [[**STD**]] and $N_{\text{eff}} = 12$ [[**L1**]]. The modular identity $\eta(-1/\tau_E) = \sqrt{-i\tau_E}\,\eta(\tau_E)$ makes $\tau_E = i$ the *unique* fixed point of the S-transformation on the imaginary axis $\tau_E = iR_\psi$, $R_\psi > 0$ [[**L0**]]. The two conditions are: (i) $\det P_{\text{eff}} = \eta^2$ [[**STD**], cited], (ii) one-loop approximation [explicit technical assumption, standard in QFT].

Implications. ψ is compact from the Scherk–Schwarz boundary condition [[**L1**]] and from the periodicity of $Z[\tau]$ [[**L0**]], independently of any geometric axiom. Scale closure $M_{\text{GUT}} = 1/R_\psi$: [[**MC**]]. $\tau = i$ is algebraically preferred for the alpha track [[**L0**]], but **alpha is NOT DERIVED** and G137-B is NOT CLOSED.

Contents

1 Motivation

Three open problems in UBT share a common root: the radius R_ψ of the ψ-circle S^1_ψ is not derived from $S[\Theta]$ without phenomenological input.

1. **Alpha (Gap G137-B)**: B_{phenom} involves $\eta(i)$, which is natural at the self-dual point $\tau_E = i$ ($R_\psi = 1$). Why is $\tau_E = i$ selected? Previously: not derived.
2. **Scale closure (EW-1b)**: $M_{\text{GUT}} = 1/R^*_\psi$; the stationarity equation for R^*_ψ required T_{kin} as external input.
3. **C2-iv**: The ψ-winding and colour fibre are decoupled in the current $S[\Theta]$. A connection might exist at $R_\psi = 1$.

This note provides two independent derivations of $R_\psi = 1$ and establishes compactness of ψ without a geometric axiom.

2 Setup: Modular Parameter and S-Transformation

Complex time $\tau = t + i\psi \in \mathbb{C}$. The Euclidean modular parameter for the ψ-circle with period $2\pi R_\psi$:

$$\tau_E = iR_\psi \in \mathbb{H}, \qquad R_\psi > 0.$$

The modular S-transformation:

$$S:\ \tau_E \longmapsto -\frac{1}{\tau_E}, \qquad \text{equivalently} \quad R_\psi \longmapsto \frac{1}{R_\psi}.$$

Definition 2.1 (Self-dual point)**.** $\tau_E = i$ ($R_\psi = 1$) satisfies $S(i) = -1/i = i$. It is the unique fixed point of S on the imaginary axis $\{iR_\psi : R_\psi > 0\}$. **[L0]**

3 Path 2: Partition Function Derivation

We present Path 2 first because it is the stronger result.

3.1 Compactness of ψ from the Scherk–Schwarz condition

Lemma 3.1 (Compactness of ψ — **[L1]**)**.** *From* `su2_twist_neff12.tex`*, the Scherk–Schwarz boundary condition on S^1_ψ is [4]:*

$$\Theta(\psi + 2\pi R_\psi) = \sigma_3\,\Theta(\psi)\,\sigma_3, \qquad \sigma_3 = \mathrm{diag}(+1, -1).$$

This condition assigns ψ a natural period $2\pi R_\psi$: diagonal components a, d are periodic ($m_n = n/R_\psi$); off-diagonal components b, c are anti-periodic ($m_n = (n + \frac{1}{2})/R_\psi$).

Consequence*: the compactness of ψ is a property of the symmetry of the field Θ, not a geometric axiom.* ***[L1]***

3.2 One-loop partition function

Lemma 3.2 (Functional determinant for a scalar on S^1 — [**STD**])**.** *For a periodic real scalar field on S^1 with modular parameter $\tau_E = iR_\psi$, the ζ-regularised one-loop functional determinant is [1, §7.2]:*

$$\det P_{\text{eff}}(\tau_E) = \eta(\tau_E)^2, \qquad \eta(\tau) = q^{1/24}\prod_{n=1}^{\infty}(1-q^n), \quad q = e^{2\pi i\tau}.$$

Hence the one-loop contribution to the effective action is $\frac{1}{2}\ln\det P_{\text{eff}} = \ln\eta(\tau_E)$.

Proposition 3.3 (UBT ψ-sector partition function — [**L1**] conditional)**.** *From Lemma 3.2 [[**STD**]] and $N_{\text{eff}} = 12$ [[**L1**], [4]], the total one-loop partition function of the UBT ψ-sector is:*

$$\boxed{Z[\tau_E] \;=\; \eta(\tau_E)^{-12}.}$$

***Derivation**: The N_{eff} effective modes (three compact phase directions $\times$ two charged fields $\times$ two Weyl helicity weights from the half-integer tower, all [[**L1**], [4]]) each contribute $\ln\eta(\tau_E)$ to $-\ln Z$:*

$$-\ln Z[\tau_E] = N_{\text{eff}}\,\ln\eta(\tau_E).$$

Explicit conditions for this result:

1. $\det P_{\text{eff}} = \eta(\tau_E)^2$ *[[**STD**]] — standard result, cited.*
2. $N_{\text{eff}} = 12$ *[[**L1**]] — proved in [4].*
3. *One-loop approximation: higher loops give $O(g^4)$ corrections that shift the value of $Z[\tau_E]$ but not the* location *of the fixed point $\tau_E = i$ (which is determined by algebra, not by the magnitude of Z).*

*Status: [**L1**] conditional on items 1–3.*

3.3 Effective compactification from periodicity of $Z[\tau]$

Proposition 3.4 (Effective compactification — [**L0**])**.** *From the standard identity [3, §9.2]:*

$$\eta(\tau_E + 1) = e^{i\pi/12}\,\eta(\tau_E).$$

For $Z[\tau_E] = \eta(\tau_E)^{-12}$:

$$Z[\tau_E + 1] = e^{-i\pi}\,Z[\tau_E] = -Z[\tau_E].$$

*The physical amplitude $|Z[\tau_E+1]|^2 = |Z[\tau_E]|^2$ is therefore periodic in $\tau_E \to \tau_E+1$, inducing the effective identification $\tau_E \sim \tau_E + 1$ as a symmetry of observables. [**L0**]*

***Consequence**: ψ is compact both from the SS condition (Lemma 3.1, [**L1**]) and from the periodicity of $Z[\tau]$ ([**L0**]), independently and consistently.*

3.4 $R_\psi = 1$ as the unique fixed point

Proposition 3.5 (Modular transformation of $S[\Theta]$: stationarity at $\tau_E = i$ — **[L0]**)**.** *From [3, §9.2]:* $\eta(-1/\tau_E) = \sqrt{-i\tau_E}\,\eta(\tau_E)$*. With* $S[\Theta] = -\ln Z[\tau_E] = 12\ln\eta(\tau_E)$*:*

$$S\left[-\frac{1}{\tau_E}\right] = S[\tau_E] + 6\ln(-i\tau_E).$$

For $\tau_E = iR_\psi$*:* $-i\tau_E = R_\psi > 0$*, so* $\ln(-i\tau_E) = \ln R_\psi$*. At* $R_\psi = 1$*:*

$$\ln(-i\cdot i) = \ln 1 = 0 \implies \frac{d}{d\epsilon}S\left[-\frac{1}{\tau_E+\epsilon}\right]\Bigg|_{\tau_E=i,\,\epsilon=0} = 0.$$

The physical action $\operatorname{Re}S[\Theta]$ *is stationary under the modular S-flow at* $\tau_E = i$*.* ***[L0]***

Theorem 3.6 ($R_\psi = 1$ from partition function symmetry — **[L1]** conditional)**.** *Under the conditions of Proposition 3.3:*

1. ***Fixed point****:* $\tau_E = i$ *satisfies* $S(\tau_E) = \tau_E$*, hence the factor* $(-i\tau_E)^{-6} = 1^{-6} = 1$ *in the S-transformation of* $Z[\tau_E]$*.* $Z[i]$ *is S-invariant.* ***[L0]***
2. ***Uniqueness on the imaginary axis****: The fixed points of* $\mathrm{SL}(2,\mathbb{Z})$ *on* $\mathbb{H}$ *are* $\tau = i$ *(fixed by* S*) and* $\tau = e^{2\pi i/3}$ *(fixed by* ST*). For* $\tau_E = iR_\psi$ *with* $R_\psi > 0$*, only* $\tau_E = i$ *is reachable. Hence* $R_\psi = 1$ *is the* unique *fixed point on the UBT modular parameter space.* ***[L0]***
3. ***Stationarity****:* $\operatorname{Re}S[\Theta]$ *is stationary under the modular S-flow at* $\tau_E = i$ *(Proposition 3.5).* ***[L0]***

Conclusion*:* $R_\psi = 1$ *is derived from the algebraic structure of* $Z[\tau_E] = \eta(\tau_E)^{-12}$ *and the modular identity for* η*. Status:* ***[L1]*** *conditional on items 1–3 of Proposition 3.3 (i.e. on* $\det P = \eta^2$ *[****[STD]****],* $N_{\text{eff}} = 12$ *[****[L1]****], and one-loop approximation).*

4 Path 1: Casimir Polynomial

This section provides an independent derivation via dynamics.

The effective action (Casimir energy on T^3 with $d = 3$, [2]; $C_{\text{tot}} = N_{\text{eff}}\cdot 3\zeta(3)/(128\pi^2) > 0$):

$$S_{\text{eff}}(R_\psi, T_{\text{kin}}) = 2\pi R_\psi\, T_{\text{kin}} + \frac{C_{\text{tot}}}{R_\psi^3}.$$

Lemma 4.1 (S-invariance condition — conditional on S-symmetry of S_{eff})**.** $S_{\text{eff}}(R) = S_{\text{eff}}(1/R)$ *if and only if:*

$$T_{\text{kin}}(R_\psi) = \frac{C_{\text{tot}}}{2\pi}\left(R_\psi^2 + 1 + R_\psi^{-2}\right). \tag{1}$$

(*Factorisation* $R^3 - R^{-3} = (R - R^{-1})(R^2 + 1 + R^{-2})$*.* ***[L0]***)

Theorem 4.2 (Polynomial fixing — **[L0]** (polynomial) + **[MC]** (S-symmetry))**.** *Substituting (1) into* $dS_{\text{eff}}/dR_\psi = 0$*:*

$$\boxed{R_\psi^6 + R_\psi^4 + R_\psi^2 = 3.} \tag{2}$$

This has the unique positive real solution $R_\psi = 1$.

Proof. Solution*:* $1 + 1 + 1 = 3$*. **[L0]**. Uniqueness:* $f(u) = u^3 + u^2 + u - 3$ *with* $u = R_\psi^2 > 0$*.* $f(1) = 0$*,* $f'(u) = 3u^2 + 2u + 1 > 0$ *for all* u *(discriminant* < 0*). Hence* $u = 1$ *is the unique positive root. **[L0]**.* N_{eff} cancels*:* $C_{\text{tot}} = N_{\text{eff}} C_{\text{unit}}$ *appears on both sides of the stationarity equation and divides out. The polynomial is independent of* N_{eff}*. Verified for* $N_{\text{eff}} \in \{1, 3, 6, 12, 24, 48\}$ *numerically.*

***Status**: the polynomial* (2) *and its unique solution are **[L0]**. The S-symmetry of* S_{eff} *(condition for Lemma 4.1) is **[MC]** — it is not derived from* $S[\Theta]$ *but is consistent with the result of Path 2 (Theorem 3.6).*

Corollary 4.3 ($R_\psi = 1$ is a minimum — **[L0]**)**.** $d^2 S_{\text{eff}}/dR_\psi^2|_{R_\psi=1} = 20 C_{\text{tot}} > 0$*. Numerically* $\approx 0.685 > 0$*. **[NUM]***

5 Comparison of the Two Paths

Path	Mechanism	Status
1 — Casimir polynomial	S-invariance of S_{eff} + stationarity $\Rightarrow R_\psi^6 + R_\psi^4 + R_\psi^2 = 3$	**[L0]** (polynomial) + **[MC]** (S-symmetry)
2 — Partition function	$Z[\tau] = \eta^{-12}$; $\tau_E = i$ is unique S-fixed point on imaginary axis	**[L1]** conditional
Agreement of two independent paths		structural evidence

The [MC] **assumption in Path 1 (S-symmetry of** S_{eff}**) is independently confirmed by Path 2**: if $Z[\tau] = \eta^{-12}$, then $S[\Theta] = -\ln Z$ satisfies $\operatorname{Re} S(-1/\tau_E) = \operatorname{Re} S(\tau_E)$ at $\tau_E = i$ (**[L0]**). Path 2 thus provides post-hoc justification for the key assumption of Path 1.

6 Implications

Scale closure (EW-1b)

$M_{\text{GUT}} = 1/R_\psi = 1$ (in natural units fixed by $R_\psi = 1$). The Weinberg angle $\sin^2\theta_W(M_Z) \approx 0.231$ via EW-1b+RG is now **[MC]** (previously blocked on R_ψ being free).

Alpha track

$\tau_E = i$ is algebraically preferred by the modular structure of $S[\Theta]$, not chosen to fit B_{phenom}. This removes the circular argument. However, Gap G137-B (deriving B_{phenom} from $S[\Theta]$) remains open. **Alpha is NOT DERIVED.**

C2-iv

At $R_\psi = 1$, the circle S^1_ψ with period 2π admits a natural $\mathbb{Z}_3$ subgroup (rotation by $2\pi/3$). Whether this $\mathbb{Z}_3$ can be identified with the colour $\mathbb{Z}_3$ (from involutions $\tau_1 \circ \tau_2 \circ \tau_3 = \text{id}$

in SU(3)) is an open question [[**MC**]]; it requires a coupling term in $S[\Theta]$ not currently present (C2-iv formal NO-GO).

Compactness of ψ

Established without axiom via two mechanisms: SS boundary condition [[**L1**]] and periodicity of $Z[\tau]$ [[**L0**]].

7 Numerical Verification

All claims verified in `tools/modular_rpsi_check.py`. Key values at $\tau_E = i$ ($q = e^{-2\pi}$):

Quantity	Value	Claim verified
$\eta(i)$	$0.768225\ldots$	—
$\eta(i)^{-12}$	$23.666602\ldots$	$Z[i]$
$\sqrt{-i \cdot i}$	1 (exact)	S-factor at $\tau_E = i$
$\ln(-i\tau_E)\|_{R_\psi=1}$	0 (exact)	stationarity
$f'(u) = 3u^2 + 2u + 1$	> 0 for all u	uniqueness of $u = 1$
$d^2S/dR^2\|_{R=1}$	$20C_{\text{tot}} \approx 0.685 > 0$	minimum
S-invariance $S_{\text{eff}}(R) = S_{\text{eff}}(1/R)$	residual $< 10^{-17}$	[**NUM**]

8 Summary

What has been derived

1. ψ **is compact**: from SS condition [[**L1**]] and from $|Z[\tau+1]|^2 = |Z[\tau]|^2$ [[**L0**]]. No geometric axiom required.

2. $Z[\tau_E] = \eta(\tau_E)^{-12}$: from $\det P = \eta^2$ [[**STD**]] and $N_{\text{eff}} = 12$ [[**L1**]]. Status: [[**L1**] conditional on one-loop approximation].

3. $\tau_E = i$ **is the unique S-fixed point on** $\{iR_\psi\}$: [[**L0**]] — algebraic fact from $S(i) = i$ and geometry of $\mathbb{H}$.

4. $R_\psi = 1$ **minimum of** S_{eff}: [[**L0**]] from polynomial (Path 1, with [**MC**] on S-symmetry of S_{eff}).

5. **Paths 1 and 2 are consistent**: Path 2 [[**L1**] cond.] post-hoc justifies the [**MC**] assumption of Path 1.

What remains open

- Derivation of $\det P = \eta^2$ directly from $S[\Theta]$ (currently [[**STD**]], cited).
- Validity of one-loop approximation (standard assumption, corrections do not shift the fixed point).

- Scale closure full proof ([[**MC**]] $\to$ [[**L1**]]): requires T_{kin} from $S[\Theta]$ without external input.
- Gap G137-B: B_{phenom} not derived. **Alpha: NOT DERIVED.**
- C2-iv: colour-ψ coupling not present in current $S[\Theta]$.

References

[1] J. Polchinski, *String Theory, Vol. 1.* Cambridge University Press, 1998.

[2] J. Ambjørn and S. Wolfram, "Properties of the vacuum. 1. Mechanical and thermodynamic," *Ann. Phys.* **147**, 1 (1983).

[3] T. M. Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed. Springer, 1990.

[4] I. D. Jaroš, "SU(2) Scherk–Schwarz Twist and $N_{\text{eff}} = 12$ in UBT," `su2_twist_neff12.tex`, 2026.